

Compiler

--- Semantics Rules

Zhang Zhizheng

seu_zzz@seu.edu.cn

School of Computer Science and Engineering,
Software College
Southeast University

Attributes of Grammar Symbols I

□ Denote the “real meaning” of grammar symbols, e.g.,

➤ $E \rightarrow E1 * \text{Digit}$ has $E.\text{val} = E1.\text{val} \times \text{Digit}.\text{val}$

$\text{Digit}.\text{type} = E1.\text{type}$

$E.\text{code} = E1.\text{code} * \text{Digit}.\text{code}$

Attributes of Grammar Symbols II

□ Two kinds of attributes for nonterminals

- **Synthesized**, whose value is defined only in terms of attributes values at the children and itself. E.g., *val*
- **Inherited**, whose value is defined only in terms of attributes values at the parents, siblings and itself. E.g., *type*

Semantics Rules I

- Specify the evaluation of the value of the attribute
- One production is attached by several semantics rules.

➤ E.g., for $E \rightarrow E1 * \text{Digit}$

$E.\text{val} = E1.\text{val} \times \text{Digit}.\text{val}$

$\text{Digit}.\text{type} = E1.\text{type}$

$E.\text{code} = E1.\text{code} \text{ '}' \text{ Digit}.\text{code}$

Semantics Rules II

- Is Activated when the host production is used in the reduction action.
- Is Used for **type checking**, **intermediate code generation**.

➤ E.g., for $E \rightarrow E1 * \text{Digit}$

$\text{Digit.type} = E1.\text{type}$

$E.\text{code} = E1.\text{code} * \text{Digit.code}$

Semantics Rules III

□ Order the evaluation of attributions I

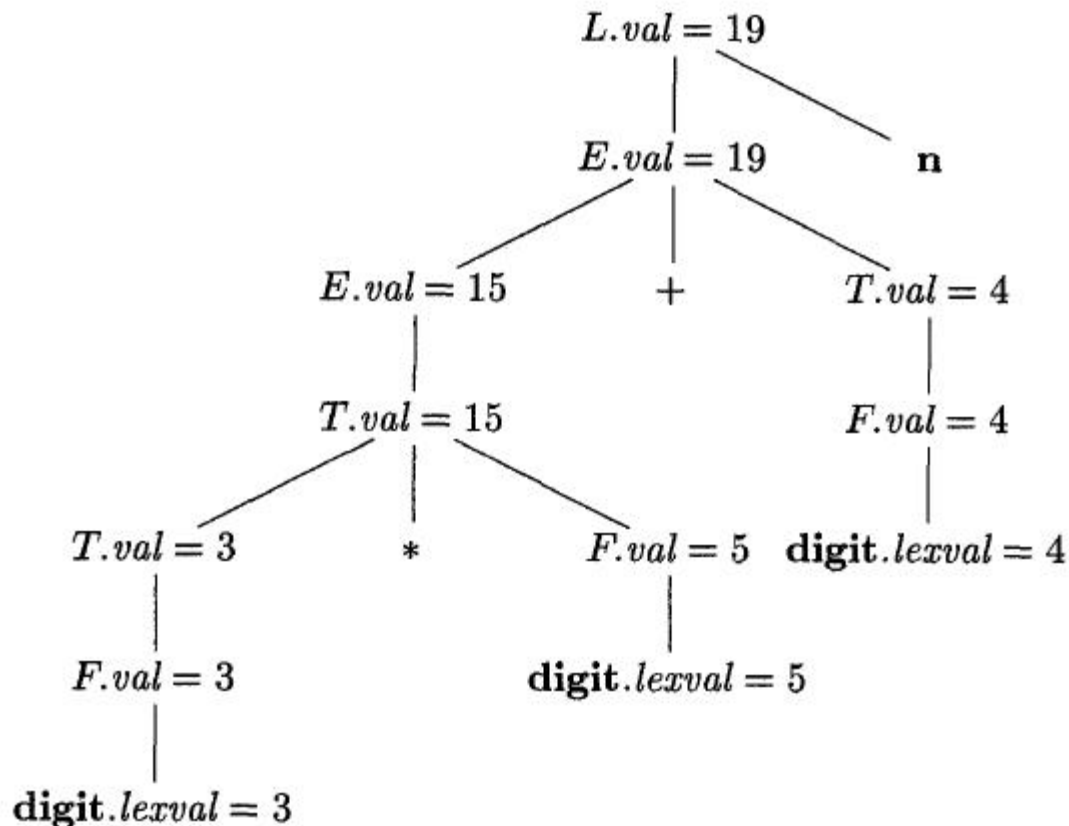
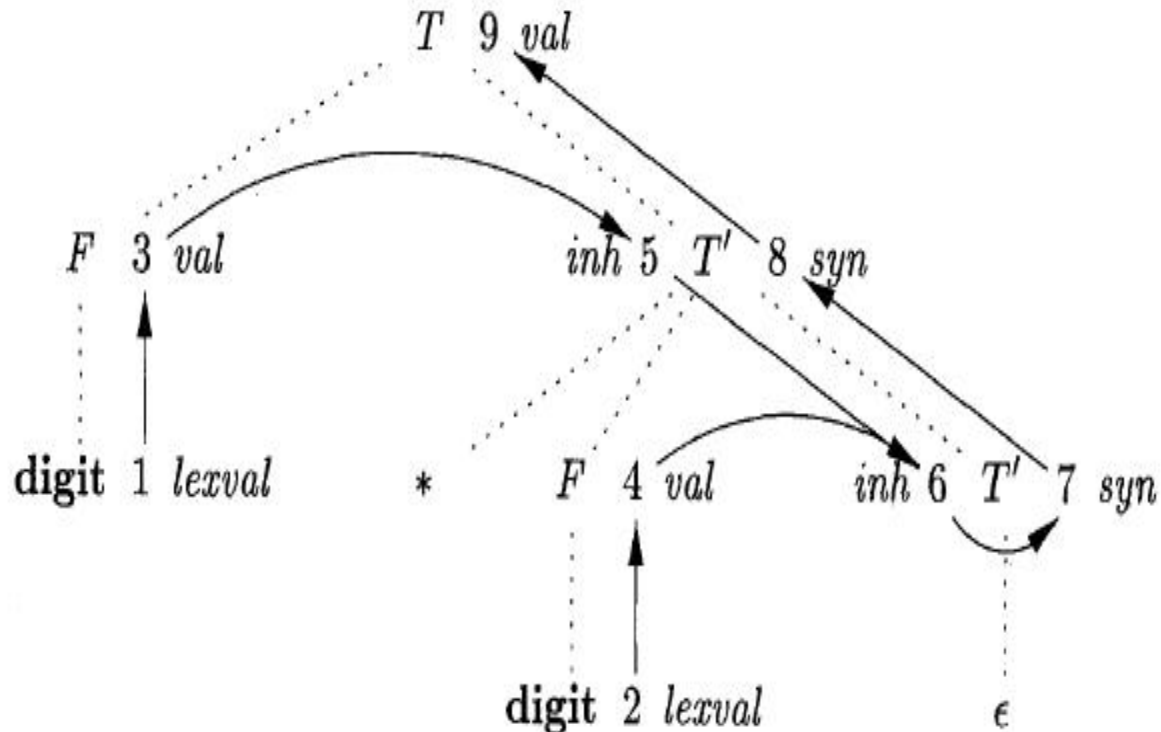


Figure 5.3: Annotated parse tree for $3 * 5 + 4 n$

Semantics Rules III

□ Order the evaluation of attributions II

PRODUCTION	SEMANTIC RULES
1) $T \rightarrow F T'$	$T'.inh = F.val$ $T.val = T'.syn$
2) $T' \rightarrow * F T'_1$	$T'_1.inh = T'.inh \times F.val$ $T'.syn = T'_1.syn$
3) $T' \rightarrow \epsilon$	$T'.syn = T'.inh$
4) $F \rightarrow \text{digit}$	$F.val = \text{digit.lexval}$



Semantics Rules III

□ Order the evaluation of attributions III

PRODUCTION	SEMANTIC RULES
1) $D \rightarrow T L$	$L.inh = T.type$
2) $T \rightarrow \text{int}$	$T.type = \text{integer}$
3) $T \rightarrow \text{float}$	$T.type = \text{float}$
4) $L \rightarrow L_1, \text{id}$	$L_1.inh = L.inh$ $\text{addType}(\text{id.entry}, L.inh)$
5) $L \rightarrow \text{id}$	$\text{addType}(\text{id.entry}, L.inh)$

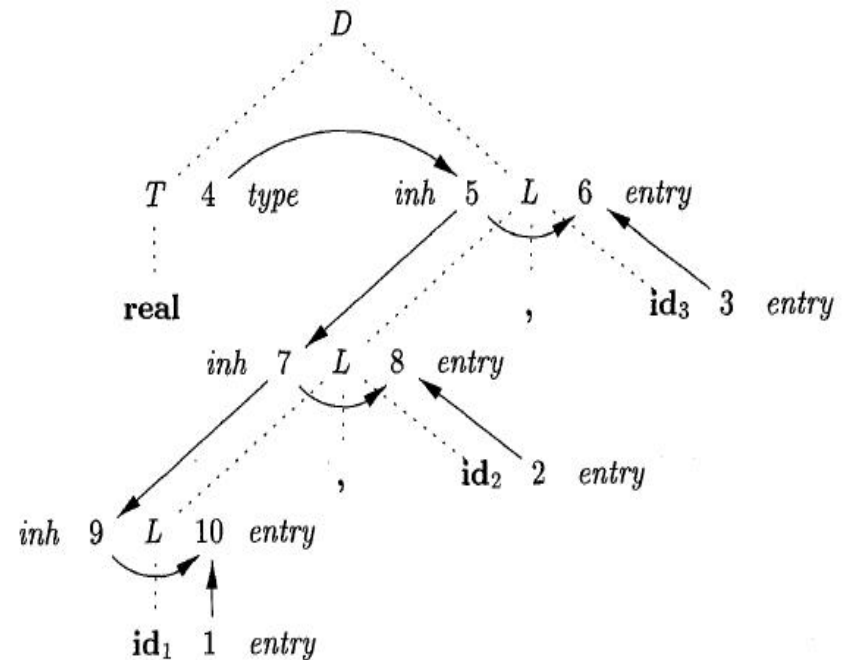


Figure 5.9: **Dependency graph** for a declaration **float id₁, id₂, id₃**

Figure 5.8: Syntax-directed definition for simple type declarations

Syntax-directed Translation

- Because the semantics rule is activated when its associated production is used. Thus, the application of the semantics rule is syntax-directed.
- However, the order of evaluation of attributes value may not align the order of applying productions in parsing. E.g.,
 - Synthesized attributes fit bottom up.
 - Inherited attributes fit top down.

Written Assignment

•Please construct **an annotated parse tree** for the input string $4+(5*6+9)*7$ where the syntax-directed definition is as following

Productions

$E \rightarrow E1 * T$

$E \rightarrow T$

$T \rightarrow T1 + F$

$T \rightarrow F$

$F \rightarrow (E)$

$F \rightarrow i$

Semantic Rules

$E.val = E1.val * T.val$

$E.val = T.val$

$T.val = T1.val + F.val$

$T.val = F.val$

$F.val = E.val$

$F.val = i.lexval$

Self Study

- Please self-study SDT and SDD according to the textbook